



An integrated optical modulator operating at cryogenic temperatures

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Photonic integrated circuits (PICs) operating at cryogenic temperatures are fundamental building blocks required to achieve scalable quantum computing and cryogenic computing technologies^{1,2}. Silicon PICs have matured for room-temperature applications, but their cryogenic performance is limited by the absence of efficient low-temperature electro-optic modulation. Here we demonstrate electro-optic switching and modulation from room temperature down to 4 K by using the Pockels effect in integrated barium titanate (BaTiO₃) devices³. We investigate the temperature dependence of the nonlinear optical properties of BaTiO₃, showing an effective Pockels coefficient of 200 pm V⁻¹ at 4 K. The fabricated devices show an electro-optic bandwidth of 30 GHz, ultralow-power tuning that is 10⁹ times more efficient than thermal tuning, and high-speed data modulation at 20 Gbps. Our results demonstrate a missing component for cryogenic PICs, removing major roadblocks for the realization of cryogenic-compatible systems in the field of quantum computing, supercomputing and sensing, and for interfacing those systems with instrumentation at room temperature.

Cryogenic technologies are becoming essential for future computing systems, a trend fuelled by the worldwide quest to develop quantum computing systems and future generations of high-performance classical computing systems^{4,5}. While most computing architectures rely solely on electronic circuits, photonic components are becoming increasingly important (Supplementary Note 1). First, photonic integrated circuits (PICs) can be used for quantum computing approaches where the quantum nature of photons is exploited as qubits^{1,2}. Second, optical interconnects can overcome limitations in bandwidth and heat leakage that are present in conventional electrical interconnect solutions for digital data transfer between cryogenic processors and the room-temperature environment⁶ (Supplementary Note 2). In addition, due to their low interaction with the environment, photons are the only viable carriers to transport quantum states over large distances. Optical interfaces are therefore essential for true quantum communication, which is necessary to connect multiple quantum computers^{7,8} and for secure remote operation of quantum computers⁹. Recently, integrated photonics have also been exploited in quantum computing architectures based on trapped ion qubits¹⁰. The scalability of such a system is directly reliant on integrated cryogenic electro-optic (EO) modulators. The need for cryogenic photonics is not limited to computing systems, but covers a wide range of technical fields, such as radio astronomy¹¹, particle physics¹² and terahertz sensing¹³.

Today, the realization of such photonic concepts is hindered by the lack of switches and modulators that operate at cryogenic temperatures with low loss, high bandwidth and low static power consumption. So far, only two concepts for cryogenic EO switches have been investigated, based either on the thermo-optic effect¹⁴ or on the plasma-dispersion effect¹⁵. Both mechanisms have physical limitations that intrinsically restrict the low-temperature performance of such devices. Thermo-optic phase shifters exploit Joule heating with large static power consumption and exhibit a bandwidth of less than a few megahertz¹⁶. Plasma-dispersion-based devices require very high doping levels to compensate for carrier freeze-out at cryogenic temperatures and are intrinsically incapable of pure phase modulation. The high doping leads to large propagation losses and devices are limited to a bandwidth of <5 GHz in microdisk modulators¹⁵. Both of these technologies are fundamentally limited from providing low-loss, low-power, high-speed switching and tuning as is fundamentally required for, for example, photonic quantum computing^{1,17}. InP-on-Si modulators have also been explored for cryogenic use¹⁸ but with limited performance and only to temperatures of 77 K, not reaching the few kelvin or below as required by cryogenic applications.

The use of EO switches based on the Pockels effect has been shown to offer low propagation losses and high bandwidth, combined with low static power consumption at room temperature^{3,19–21}. While the Pockels effect has no intrinsic physical limitations for use at cryogenic temperature²², making a Pockels device requires an integrated material that retains a large Pockels coefficient and that does not suffer from additional spurious effects at low temperature. No integrated cryogenic Pockels modulator has previously been reported, but room-temperature devices have recently been demonstrated using organics²³, PbZr_{1-x}Ti_xO₃ (ref. ¹⁹), LiNbO₃ (ref. ²⁰) and BaTiO₃ (ref. ³). Among them, BaTiO₃ stands out due to it having the largest Pockels coefficients³ and exhibiting compatibility with advanced silicon photonics platforms²⁴. We complete this triumvirate by demonstrating that BaTiO₃ is also an ideal candidate for cryogenic EO integration.

Both the nonlinear optical (NLO) properties and structural behaviour of BaTiO₃ thin films are entirely unknown at temperatures below 300 K. In fact, even in bulk BaTiO₃ crystals the NLO behaviour below 270 K is not known, and the room-temperature NLO behaviour of BaTiO₃ thin films has only recently been thoroughly investigated³. The phase transitions of thin films are expected to differ from bulk crystals²⁵ due to the structural mismatch and thermal stress that exists between the substrate and the BaTiO₃ layer^{26,27}. They are entirely unknown for BaTiO₃ on Si.

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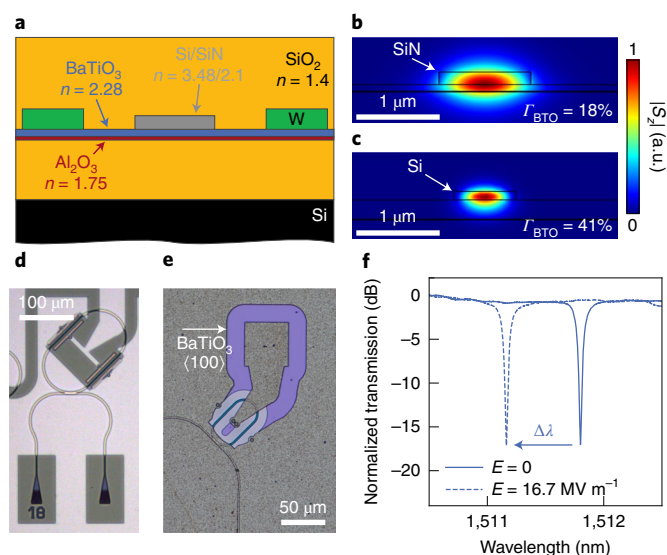


Fig. 1 | BaTiO₃ EO device concept. **a**, Schematic cross-section of the devices. A Si or SiN forms a strip waveguide (grey) on top of an BaTiO₃ layer (blue). A thin Al₂O₃ layer (red) was used to improve adhesion (Methods). Lateral electrodes fabricated with W (green) are used to apply an electric field across the BaTiO₃. The devices are embedded in SiO₂ (yellow) layers on top of silicon substrates (black). The refractive indices of BaTiO₃ and SiN were measured using spectroscopic ellipsometry. **b,c**, Simulations of the Pointing vector, $|S_z|$, of the transverse electric mode of the SiN waveguide geometry (**b**) and the Si waveguide geometry (**c**). The two waveguide geometries show an optical confinement in BaTiO₃, Γ_{BTO} of 18% and 41%, respectively. **d,e**, Optical micrographs of BaTiO₃-SiN (**d**) and BaTiO₃-Si (**e**) racetrack resonators used to characterize BaTiO₃ and demonstrate device functionality. The phase-shifter section is embedded in the resonator, which is evanescently coupled to access waveguides. The signal electrodes are connected to pads and the ground electrodes to a ground plane, for contacting using electrical probes. The number 18 in **d** is a chip layout reference. **f**, Characterization principle of resonant EO switches, showing example data of the shift in resonance wavelength between no applied field (solid line) and 16.7 MV m⁻¹ (dashed line), which in this device corresponds to 150 V resulting in a shift of 4.3 pm V⁻¹. The shift in resonance wavelength, $\Delta\lambda$, is measured for an applied electric field and converted to the material properties of BaTiO₃ (Methods).

Predictions of the Pockels tensor at cryogenic temperatures based on data at higher temperatures is not possible because in complex oxide materials the functional properties can change drastically with temperature. Here we determine the cryogenic behaviour of BaTiO₃ thin films by analysing the performance of BaTiO₃-based EO switches at temperatures down to 4 K. Our results show that efficient EO switching at cryogenic temperature is indeed possible and with bandwidths beyond 30 GHz. We also demonstrate the applicability of such devices for low-power switching and tuning as well as high-speed data modulation at 20 Gbps at 4 K.

Here we use two waveguide designs (Fig. 1a) fabricated on single-crystalline BaTiO₃ layers bonded to SiO₂-buffered silicon substrates (Methods). In the first design, SiN-based waveguides (Fig. 1d) allowed us to study the pure NLO properties of BaTiO₃ in absence of mobile charge carriers that could result in an additional, non-Pockels EO response. In the second design, Si waveguides served as more efficient devices (Fig. 1e) to demonstrate high-speed data modulation. The enhanced efficiency originates from a larger optical-mode overlap with the BaTiO₃ layer (41%; Fig. 1c) than with the SiN waveguides (18%; Fig. 1b). We found that the propagation losses (5.6 dB cm⁻¹, SiN device) were not affected by the presence

of BaTiO₃ in the active section (Supplementary Note 3) throughout the temperature range studied.

To characterize the NLO behaviour of BaTiO₃ at 4 K, we measured the induced resonance shift in a racetrack resonator as a function of the d.c. bias (Fig. 1f), from which we determined the refractive index change of BaTiO₃ (Δn_{BTO}) as a function of the applied electric field (Methods). This dependence allows us to study two of the three expected features of Pockels-based switching³: NLO hysteresis and angular anisotropy, the third being the persistence of the Pockels effect at high frequencies (>10 GHz)³.

The NLO response with a hysteretic behaviour (Fig. 2a) indicates that a non-vanishing Pockels effect is preserved in BaTiO₃ down to a temperature of 4 K. We determine the effective Pockels coefficient, r_{eff} , by analysing the hysteretic behaviour of the refractive index change (Supplementary Note 4). The dependence of r_{eff} on device orientation (Fig. 2b) reveals the second signature of the Pockels effect, its anisotropy. The reduced magnitude at 4 K compared with room temperature is due to a temperature dependence of the Pockels effect, as discussed below. While r_{eff} is reduced with temperature, the EO response is expected to be present at high frequencies also at low temperature. Indeed, we observe a constant EO response in racetrack resonators with a low quality factor ($Q \approx 1,800$) up to 30 GHz (Fig. 2b). The frequency response is expected to remain flat at even higher frequency but could not be measured in our experiment (Methods). The hysteretic behaviour, anisotropy and high-speed response prove the presence of the Pockels effect in BaTiO₃ at 4 K.

We performed electrical characterization of the material at low temperature using dedicated electrical test structures (Supplementary Note 5). The resistivity of BaTiO₃ at 4 K is very high, >10⁹ Ω m. In fact, the measured current is dominated by capacitive charging and ferroelectric switching currents (Fig. 2d). The field-dependent capacitance shows clear hysteretic characteristics (Fig. 2e), consistent with ferroelectric domain switching.

The measured r_{eff} at 4 K is lower than that at room temperature (Fig. 2b), which has two causes. First, the Pockels effect itself is generally temperature dependent due to changes in strain and polarization of the crystal²⁸. Second, the non-zero elements of the Pockels tensor depend on the crystal symmetry, which can change abruptly with temperature due to structural phase transitions. BaTiO₃ bulk crystals are known to transition from a tetragonal phase at room temperature to orthorhombic and rhombohedral phases at lower temperatures (~270 K and ~200 K, respectively)²⁵. Such transitions change the elements of the Pockels tensor and modify the magnitude of the effective Pockels coefficients²⁸. Because phase transitions of thin-film materials can be drastically affected by substrate strain^{26,27,29}, studying the properties of thin-film BaTiO₃ becomes critical when considering cryogenic applications. To investigate the effects of possible phase transitions, we measured r_{eff} in a range from 4 to 340 K. Indeed, the magnitude of r_{eff} is strongly temperature dependent (Fig. 3). A peak around 240 K, with $r_{\text{eff}} > 700$ pm V⁻¹, is consistent with the reported divergence of the r_{42} element of the Pockels tensor close to the tetragonal-orthorhombic transition²⁸. Consistently, the permittivity of the BaTiO₃ layer (Methods) also shows a peak in the same temperature range (Supplementary Note 6), confirming that the abrupt change in r_{eff} is caused by a phase transition. Below 240 K, the magnitude of r_{eff} decreases gradually to around 200 pm V⁻¹ at 4 K. In addition to the phase transition at 240 K, a second phase transition occurs above 100 K causing a rapid change in r_{eff} of 90° devices. This phase transition is also observed in the qualitative behaviour of the NLO hysteresis, which shows that the transition is induced by the electric field (Supplementary Note 6). While r_{eff} of BaTiO₃ is reduced at 4 K compared with that at room temperature, the value of ~200 pm V⁻¹ is still larger than most other material systems at room temperature^{19,20}. The effect of a reduced Pockels coefficient on the energy efficiency of EO switching is partially compensated for by a simultaneous reduction of

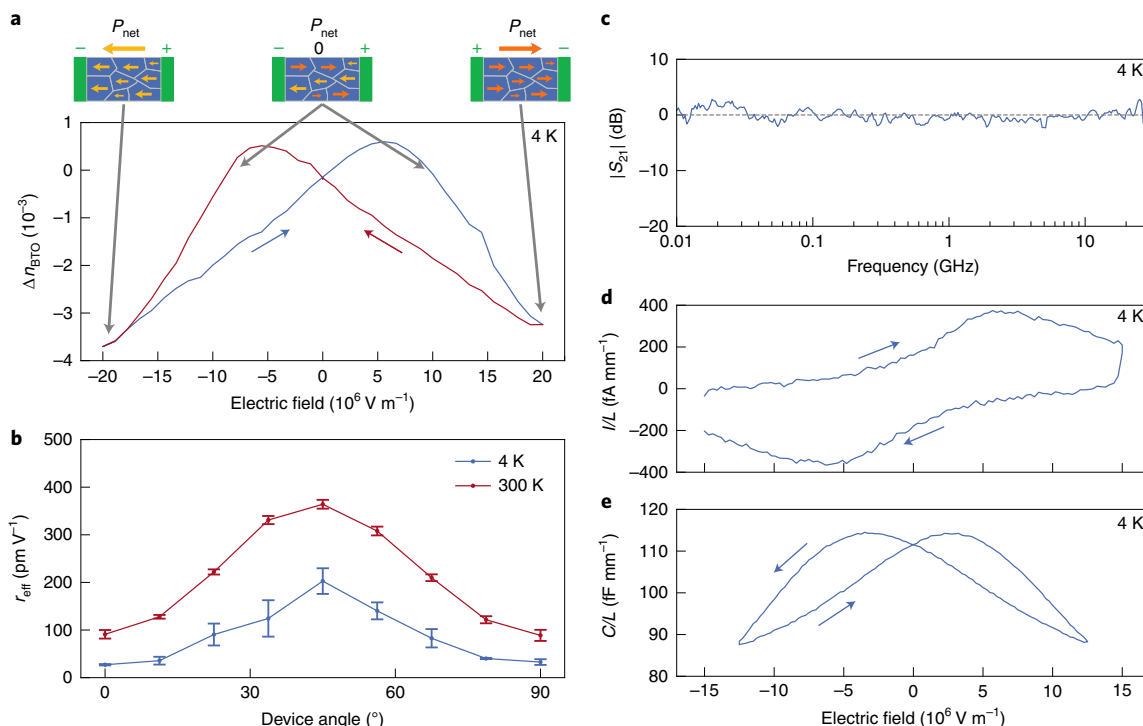


Fig. 2 | EO and electrical response of BaTiO₃-based optical switches at 4 K. **a**, Refractive index change of BaTiO₃ as a function of applied electric field for a device in the 11.25° direction (as defined in **b**). The hysteretic behaviour between increasing (blue) and decreasing (red) voltage originates from ferroelectric domain switching in the BaTiO₃, as illustrated schematically on top. P_{net} is the net polarization of all the domains. The arrows indicate the polarization of individual ferroelectric domains (blue) when a field is applied to lateral electrodes (green). **b**, Angular anisotropy of the effective Pockels coefficient in BaTiO₃ measured at 4 K and 300 K. The angle is defined relative to the BaTiO₃ <100> direction. The same anisotropy as for BaTiO₃ at room temperature is observed but with reduced magnitude. The error bars show the combined standard error of the fit and from averaging measurements of multiple devices with the same orientation. **c**, EO S_{21} parameter of BaTiO₃ ring resonator showing a flat response up to a frequency of 30 GHz at 4 K. The dashed line indicates 0 dB as a guide to the eye. **d**, Current (I) measured as a function of electric field across the BaTiO₃ layer showing extremely low current flowing through the material. The current is dominated by capacitive charging, causing the offset between the sweep directions (indicated by the arrows), together with ferroelectric switching current resulting in the observed peaks (Supplementary Note 5). **e**, Capacitance (C) as a function of electric field, showing characteristic ferroelectric hysteresis and field-dependent permittivity.

the permittivity of BaTiO₃ (Supplementary Note 6). In addition, the conductivity of BaTiO₃ is reduced by more than four orders of magnitude (Supplementary Note 5), resulting in a negligible static power consumption of BaTiO₃ devices in cryogenic environments. No material instability or drift, as could, for example, be caused by pyroelectric effects, were observed at cryogenic or room temperature in any experiment.

We have reported the measurement of the cryogenic NLO properties of BaTiO₃. The methods used to characterize these properties will enable further research to improve the cryogenic performance through engineering of the material properties²⁹. However, already today we can use BaTiO₃ to demonstrate cryogenic devices with outstanding performance.

We demonstrate the applicability of BaTiO₃ for cryogenic photonic applications by two examples: low-power EO switching and high-speed data modulation. For switching we use a 500 μm Mach-Zehnder interferometer with 2 × 2 multimode interference splitters, applying a voltage to one arm. Because the leakage current through BaTiO₃ at 4 K is 10⁴ times lower than at 300 K, less than 10 pW static power is consumed when inducing a π phase shift to switch between the two optical outputs using an electric field of $6 \times 10^6 \text{ V m}^{-1}$ (Fig. 4a,b), corresponding to a voltage of ~50 V, and a voltage-length product, $V_{\text{eff}}L$, of 5 V cm, in the given device geometry. Compared with state-of-the-art technology based on thermo-optic phase shifters¹⁴, static tuning using BaTiO₃ is one billion times more power

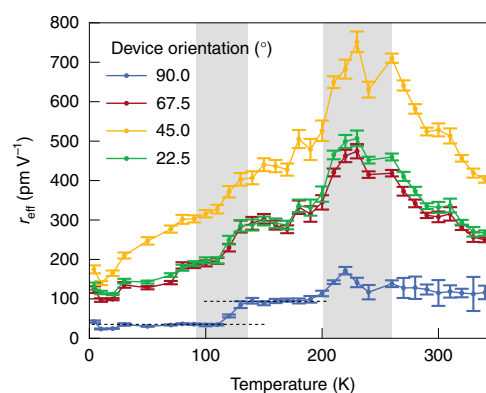


Fig. 3 | Temperature dependence of the Pockels effect in BaTiO₃.

The effective Pockels coefficient along different crystal orientations at temperatures from 4 K to 340 K. The peak around 240 K is the signature of a phase transition in BaTiO₃. A second, field-induced phase transition occurs around 100 K, causing a sharp drop of r_{eff} in 90° devices (indicated by horizontal dashed lines). This phase transition is also evident in the qualitative evaluation of the optical response (Supplementary Note 6). The grey areas indicate the temperature ranges of the respective phase transitions. The error bars show the standard error of the fit used to extract the Pockels coefficients (Supplementary Note 4).

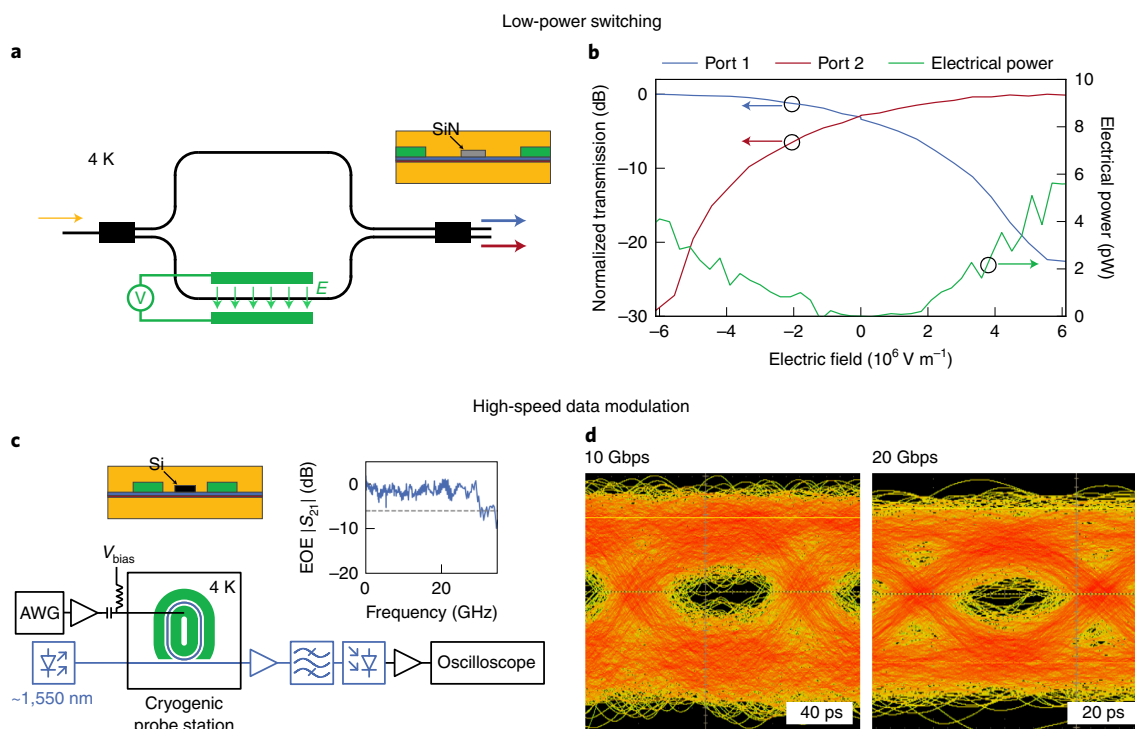


Fig. 4 | Demonstration of low-power switching and high-speed data modulation with BaTiO₃-based devices at 4 K. **a**, Schematic of Mach-Zehnder configuration used to switch between two ports. The yellow arrow indicates the input port and the red and blue arrows indicate output ports 1 and 2, respectively. Multimode interference splitters were used to split and combine the signal at the input and output ports. A voltage source, V , is used to apply an electric field, E , indicated by green arrows, across one arm of the Mach-Zehnder interferometer. The inset shows the waveguide cross-section. **b**, Transmission from both ports of a Mach-Zehnder switch as a function of applied electric field, along with the static power consumption. When fully switching between outputs, less than 10 pW static power is consumed, and only 30 pJ of dynamic energy. The field is kept below the coercive field of BaTiO₃ (Supplementary Note 9) to exclude contributions from ferroelectric domain switching. **c**, Schematic of the experimental setup for data modulation. The data signal was generated using an arbitrary waveform generator (AWG), which was amplified and then combined with a bias voltage, V_{bias} , using a bias tee. A tunable laser set to $\sim 1,550$ nm provided the optical carrier. After modulation in the cryogenic probe station, the optical signal was amplified by a fibre amplifier and filtered before being detected by a photodiode. The electrical signal from the photodiode was amplified and then recorded on a real-time oscilloscope. The left inset shows the waveguide cross-section and the right inset shows the electro-optic-electric (EOE) frequency response of the modulator. **d**, Eye diagrams recorded at 10 and 20 Gbps with $V_{\text{pp}} = 1.7$ V, corresponding to a modulation energy of 45 fJ bit⁻¹. The opening of the eyes is limited by noise from amplified spontaneous emission of the amplifier used in the experiment (Supplementary Note 8).

efficient. The dynamic energy of the switch is ~ 30 pJ, which could be reduced to ~ 2 pJ, at a voltage of < 5 V, in an optimized device geometry (Supplementary Note 7). Another important metric for EO switches is the losses in the device. On the basis of the measured propagation losses (5.6 dB cm^{-1}) and the device geometry, we estimate an insertion loss of < 1 dB (Supplementary Note 3). The propagation losses are dominated by scattering losses in the SiN waveguide, which can be reduced substantially by using a state-of-the-art fabrication process. SiN waveguides with propagation losses as low as 1 dB m^{-1} have been reported³⁰.

As a second example, we performed data modulation experiments by sending a pseudo-random bit sequence to a ring modulator ($Q \approx 6,000$) fabricated with BaTiO₃-Si waveguides and recording the optical eye-diagram (Fig. 4c,d). Data transmission at rates up to 20 Gbps are achieved with our experimental setup using a drive voltage (V_{pp}) of just 1.7 V, resulting in an extremely low energy consumption of 45 fJ bit⁻¹.

In conclusion, we have shown that BaTiO₃ thin films can be used to realize EO switches and modulators for efficient cryogenic operation of Si PICs. We have demonstrated low-power switching, as well as high-speed data modulation. Combining BaTiO₃ with Si PICs, we make a building block available that was previously inaccessible for any cryogenic circuit. We anticipate that these components are a

milestone for a versatile platform of cryogenic photonics for applications as diverse as quantum computing and communication^{1,8}, astronomy¹¹, fundamental physics¹² and cryogenic sensing concepts¹³.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41563-020-0725-5>.

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Methods

Device design and fabrication. Single-crystalline BaTiO₃ was deposited on top of an epitaxial 4-nm-thick SrTiO₃ seed layer by molecular beam epitaxy on 8" silicon-on-insulator wafers with a 220-nm-thick device silicon layer for SiN-based devices, and on 2" silicon-on-insulator wafers with a 100-nm-thick device silicon for Si-based devices, following a process described elsewhere³. Direct wafer bonding was used to transfer the BaTiO₃ and device Si layers onto high-resistivity Si wafers capped with a 3-μm-thick thermal oxide. The specifics of the direct wafer bonding process can be found in ref. ³.

For SiN-based waveguides, the device Si layer was removed by dry etching, followed by chemical vapour deposition of SiN. The waveguide layer (Si or SiN) was patterned by dry etching. After waveguide patterning, a combination of SiO₂ cladding deposition, via etching, and metallization was used to form the final cross-section. Intermediate annealing steps at 400 °C under O₂ atmosphere were used to reduce propagation losses²¹.

The SiN-based waveguides use an 80-nm-thick BaTiO₃ layer and a 150-nm-thick SiN layer. The strip-waveguide width is 1.1 μm. The SiN thickness and waveguide width were chosen to ensure guiding of a single transverse electric mode, and to maximize the overlap of that mode with the BaTiO₃ layer. The electrode-to-electrode gap is 9 μm to ensure no added losses caused by metal absorption. The racetrack resonators have a bend radius of 50 μm and straight sections of 75 μm length. The coupling gap (0.53 μm) between the access waveguide and the resonator was optimized for critical coupling.

The Si strip waveguides were fabricated using 225-nm-thick BaTiO₃ and 100-nm-thick Si. The waveguide width is 0.75 μm. As for the SiN-based waveguides, the waveguide dimension, including the Si thickness, were designed for transverse electric single-mode operation, with an optimized optical overlap with the 225-nm-thick BaTiO₃ layer. The racetrack resonator that was used has a bending radius of 15 μm and straight sections of 30 μm, and the electrode-to-electrode gap is 2.3 μm. The small electrode gap causes additional propagation losses in the resonator ensuring a *Q*-factor allowing >30 GHz bandwidth. The coupling gap of ~0.1 μm was optimized for critical coupling.

Cryogenic measurements. The cryogenic EO measurements were performed in a Lakeshore CPX cryogenic probe station, fitted with radiofrequency (40 GHz BW, K-type connectors) and optical feedthroughs. Direct current and radiofrequency signals were applied to the devices using radiofrequency probes, and optical coupling was achieved using a fibre array with polarization maintaining fibres for 1,550 nm. A tunable laser (EXFO T100S-HP) and power meter were used to record transmission spectra (EXFO CT440). The cryogenic electrical measurements were performed in a Janis cryogenic probe station equipped with d.c. probes. Current–voltage and capacitance–voltage measurements were performed using a parameter analyser. Both cryogenic probe stations were cooled by liquid helium to a base temperature of 4.2 K.

Direct current EO characterization. The d.c. EO response was extracted by applying a voltage to the electrodes of a racetrack resonator and measuring the shift in resonance wavelength ($\Delta\lambda$), compared with the unbiased case, as a function of the applied voltage. From the measured wavelength shift, the change in BaTiO₃ refractive index (Δn_{BTO}) can be estimated as

$$\Delta n_{\text{BTO}} = \frac{\lambda_0 \times \Delta\lambda}{\text{FSR} \times L_{\text{E}} \times \Gamma_{\text{BTO}}}$$

where Γ_{BTO} is the optical confinement in BaTiO₃, FSR is the free spectral range of the resonator, L_{E} is the electrode length and λ_0 is the resonance wavelength with no voltage³. The effective Pockels coefficient, r_{eff} , was then determined according to the procedure described in Supplementary Note 4.

Radiofrequency response. To measure the EO frequency response (EO S_{21}) a vector network analyser (VNA, Keysight PNA 50 GHz) was used to apply the

electrical stimulus to a BaTiO₃ ring modulator. The modulated optical signal was applied to a photodiode (Newport 1024) and the response recorded by the VNA. Electrical calibration was performed before the measurement, and the response of the photodetector was compensated for the data analysis. While the VNA could generate signals up to 50 GHz, the bandwidth of the photodetector was 26 GHz, which in combination with large frequency-dependent electrical losses in the cryogenic probe station (Supplementary Note 8) makes it impossible to measure the bandwidth beyond 30 GHz.

Data modulation experiments. For the data modulation experiment, a racetrack resonator modulator with a Si strip-waveguide was used. The electrical signal was generated using an arbitrary waveform generator. A pseudo-random bit stream of $2^7 - 1$ bits was used for modulation. The electrical signal was pre-distorted to compensate for the finite time-response of the electrical signal path (Supplementary Note 8). The signal was amplified using a radiofrequency amplifier and sent to the cryogenic setup, with an estimated voltage swing on the device of 1.7 V (Supplementary Note 8). A Pritel FA-23 fibre amplifier was used to amplify the modulated optical signal which was applied to a photodiode and recorded on an oscilloscope.

Data availability

The data from the electro-optic measurements analysed and presented in Figs. 2a–c, 3 and 4 are available in an online repository with the digital object identifier: <https://doi.org/10.5523/bris.3dwfddsnkun32ghpq4zuxft11>. The remaining data supporting the findings of this study are available from the corresponding author upon reasonable request.

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Author contributions

F.E. and J.F. fabricated and structurally characterized the epitaxial BaTiO₃/SrTiO₃ layer stack with support from H.S. F.E. designed all photonic circuits and fabricated them with support from D.C. F.E., P.S. and S.A. performed optical simulations for the device design. G.E.V.-G., F.E., A.A.G., A.H., G.D.M. and J.B. characterized the electro-optic performance at different temperatures, including low-speed and radiofrequency measurements. The electro-optic data were analysed by F.E. and A.A.G. F.E. and P.S. performed all electrical measurements. The concept for this work was defined by S.A., G.D.M. and M.G.T. and implemented by F.E. with support of S.A. and J.B. F.E., J.F. and S.A. wrote the manuscript with contributions from all authors.

Competing interests

F.E., J.F. and S.A. are involved in commercially developing barium titanate technologies at Lumiphase AG. M.G.T. is involved in developing photonic quantum technologies at PsiQuantum Corporation.

Additional information

Supplementary information is available for this paper at <https://doi.org/10.1038/s41563-020-0725-5>.

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